

## Project Design Phase-I

### Solution Architecture

|               |                                        |
|---------------|----------------------------------------|
| Date          | June 2026                              |
| Project Name  | AI-Enhanced Intrusion Detection System |
| Maximum Marks | As Applicable                          |

#### Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. The AI-Enhanced IDS architecture consists of the following layers:

#### Architecture Diagram Description:

Network Traffic (Raw Packets) → Data Preprocessing (NSL-KDD Feature Extraction) → ML Model (Random Forest Classifier) → Flask REST API (/predict endpoint) → SOC Dashboard (HTML/Chart.js) → Security Analyst

#### Data Flow:

1. Network traffic data is captured and converted into NSL-KDD feature format (41 features).
2. Features are preprocessed: label encoding for categorical fields, standard scaling for numeric fields.
3. The trained Random Forest model classifies traffic as: Normal, DoS, Probe, R2L, or U2R.
4. The Flask API receives feature input via POST /predict and returns the predicted attack class with confidence score.
5. The real-time dashboard polls the API and displays threat counts, charts, and a live alert feed.

#### Component Table:

| S.No. | Component                    | Description                                                          | Technology                           |
|-------|------------------------------|----------------------------------------------------------------------|--------------------------------------|
| 1.    | User Interface               | SOC analyst views real-time threat dashboard, charts, and alert feed | HTML5, CSS3, JavaScript, Chart.js    |
| 2.    | Application Logic - Frontend | Dashboard polling, alert rendering, data visualization               | Vanilla JS, Fetch API, Chart.js      |
| 3.    | Application Logic - Backend  | REST API to serve ML predictions and threat logs                     | Python 3.11, Flask, Flask-CORS       |
| 4.    | Machine Learning Model       | Random Forest classifier for 5-class network traffic classification  | Scikit-learn, Pandas, Joblib (.pkl)  |
| 5.    | Data Preprocessing           | Feature encoding, scaling, and transformation of NSL-KDD features    | Pandas, LabelEncoder, StandardScaler |
| 6.    | Dataset / File Storage       | NSL-KDD benchmark dataset (CSV) for model training and evaluation    | Local filesystem / CSV files         |

|    |                            |                                                  |                                             |
|----|----------------------------|--------------------------------------------------|---------------------------------------------|
| 7. | Infrastructure<br>(Server) | Local development server, deployable<br>to cloud | Python venv, Flask dev server /<br>Gunicorn |
|----|----------------------------|--------------------------------------------------|---------------------------------------------|